Rare Is Not Never

AP Statistics · Unit 2 · Topic 2.10 · B: 2026-11-05 · E: 2026-11-25 · TEACHER KEY

CED TETHER

- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **EU UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **LO UNC-3.A** — Estimate probabilities of binomial random variables using data from a simulation. [Skill 3.A]
- **EK UNC-3.A.1** — A probability distribution can be constructed using the rules of probability or estimated with a simulation using random number generators.
- **EK UNC-3.A.2** — A binomial random variable, X, counts the number of successes in n repeated independent trials, each trial having two possible outcomes (success or failure), with the probability of success p and the probability of failure 1 - p.
- **LO UNC-3.B** — Calculate probabilities for a binomial distribution. [Skill 3.A]
- **EK UNC-3.B.1** — The probability that a binomial random variable, X, has exactly x successes for n independent trials, when the probability of success is p, is calculated as P(X = x) = C(n, x) · p^x · (1 - p)^(n - x), for x = 0, 1, 2, . . . , n. This is the binomial probability function.

Essential question: When does a low binomial probability count against a familiar chance model without proving it false?

DOK-3 skill: 4.B · **FRQ pattern:** `binomial-conditions-then-is-this-outcome-unusual`

DOK rationale: Part (c) coordinates assumptions about repeated shots with a cumulative tail probability, adjudicates two overconfident claims, and identifies new evidence that would change the conclusion.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Record Claim A, Claim B, and neither votes without resolving the tail probability.
- Begin the combined 4.10–4.12 video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and separate model fit, probability, and conclusion.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Commit to one claim or neither and cite the result or graph before the lesson.
- Complete the follow-along about BINS and exact versus cumulative binomial probability.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- What values belong in the event 3 or fewer?
- Which BINS condition is an assumption about consecutive free throws rather than a stated fact?
- Does 0.0106 mean impossible, ordinary, or something between those?

- What pattern in a larger set of comparable shots would update your conclusion?

Adult role
Solo teacher. Circulate 5 pairs. Require a condition check and cumulative event before accepting language about a slump or chance variation.

[WARN] WATCH FOR

- Using only the probability of exactly 3 for a 3-or-fewer claim.
- Listing BINS without assessing independence or constant probability in context.
- Treating one unusual set as proof of a changed parameter.

SCORING PART (c) — E / P / I

Expected elements

- **checks-bins-with-assumption** (required) — Checks binary outcomes, independence, fixed n, and constant p while treating independence and constant probability as assumptions to assess
- **computes-lower-tail** (required) — Computes or reports $P(X \leq 3) \approx 0.0106$
- **adjudicates-both-absolutes** (required) — Explains that the result is unusual, not ordinary, but remains possible and does not by itself prove a changed shooting rate
- **names-informative-new-evidence** (required) — Uses a larger set of comparable future attempts and explains how its rate would update the conclusion

E	Checks all four conditions with a qualified independence judgment, computes the lower-tail probability, rejects both overstatements for distinct reasons, and describes larger-sample evidence that could update the conclusion.
P	Gets the probability and unusualness judgment but incompletely checks conditions, treats one claim as wholly correct, or names more data without explaining what pattern would matter.
I	Uses only $P(X = 3)$ for the lower tail, calls the outcome impossible or ordinary without numerical support, or ignores whether the binomial model applies.

Common mistakes

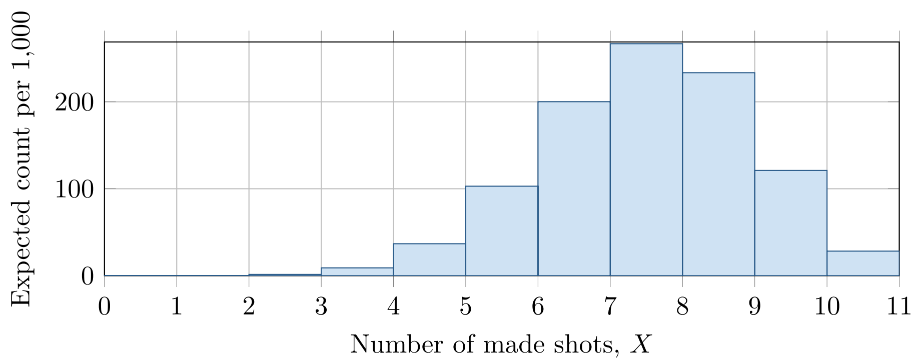
- Computing only $P(X = 3)$ when the evidence is 3 or fewer
- Calling shots independent merely because there are 10 of them
- Treating a small probability as proof that the career probability changed
- Saying any result from 10 attempts must be ordinary because the sample is small

[STAR] TODAY'S PROBLEM — annotated

Suppose a basketball player has made 70 percent of her free throws over a long career. Tonight she makes 3 of 10 attempts. Let X be the number she would make in 10 attempts if her chance on each shot were still 0.70. The graph shows the binomial model as expected frequencies per 1,000 sets of 10 shots.

Claim A: "She is in a slump—making only 3 of 10 never happens at her career rate."

Claim B: "With only 10 shots, 3 makes is ordinary chance variation."



FIRST TAKE (5 min)

Which claim, if either, do you trust more? Cite one detail from the result or graph.
Key: Not graded. Preserve the possibility of neither claim; both statements intentionally overreach in different directions.

Rules callout "BINS BEFORE BINOMIAL (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Define the random variable X in context, state n and p, and write a probability expression for exactly 3 makes.

Key: X is the number of free throws the player makes in 10 attempts under her career-rate model. Thus $n = 10$ and $p = 0.70$, and the requested expression is $P(X = 3) = \binom{10}{3}(0.70)^3(0.30)^7$.

DOK 2 **(b)** Calculate $P(X = 3)$ using the binomial probability formula and interpret the result in context.

Key: $P(X = 3) = \binom{10}{3}(0.70)^3(0.30)^7 = 0.009001692 \approx 0.0090$. If her make probability remains 0.70 with independent shots, only about 0.9 percent of 10-shot sets contain exactly 3 makes.

DOK 3 **(c)** Check all four binomial conditions for this model, explicitly assessing whether independence is plausible. Then compute $P(X \leq 3)$ and adjudicate Claims A and B: which claim, if either, is warranted, and why? State what additional shooting evidence would change your conclusion.

Key: Binary outcomes are make or miss; there are 10 attempts; the model uses the same $p = 0.70$; and it assumes independence. The last two are plausible only for a stable session without fatigue, injury, or carryover between shots. Under the model, $P(X \leq 3) = \sum_{k=0}^{3} \binom{10}{k}(0.70)^k(0.30)^{10-k} = 0.0105920784 \approx 0.0106$. Neither claim is fully warranted: Claim B wrongly calls this unusual result ordinary, while Claim A wrongly calls it impossible and treats one short set as proof that the rate changed. Many comparable future shots near 70 percent would favor chance variation; a sustained rate well below 70 percent would strengthen the slump claim.

EXIT

One thing the video changed about my first take: ______________________________